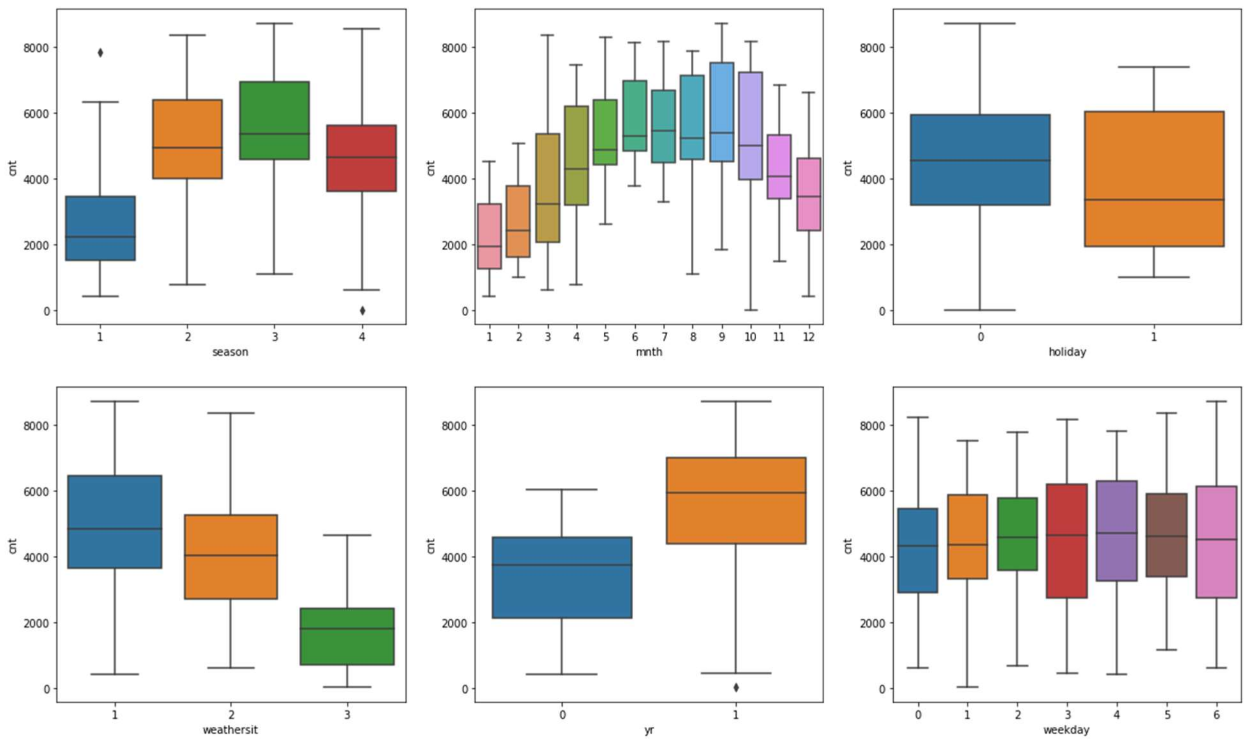
Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

After going through the dataset, following columns are identified as categorical variables.

1. season : From box plot its easily inferred that spring has less effect on the count of the bike rental. However, Fall and Summer has higher counts of bikes respectively. Next season impacted being Winter. 1:'Spring',2:'Summer',3:'Fall',4:'Winter'
2. mnth : September and October month have see highest number of bike rentals.
3. holiday : Median of rentals on holiday is significantly less than that during non-holidays. Hence rentals on holiday is less.
4. weathersit : On the days of “ Light Snow or Rain” less rentals are see, on extreme weather conditions like “Heavy Rain” no record of bike rentals. However, during Clear, Mist or cloudy days has seen good rentals.
5. yr : Year on Year bike rental demand is increasing.
6. weekday : The average rentals on each day is similar with less variations. Mondays and Fridays has seen large distribution of rentals.
7. workingday : Friday and Monday has seen high distribution of rentals.



2. Why is it important to use drop_first=True during dummy variable creation?

As part of dummy variable creation from categorical variable, we convert each value in categorical variable into separate columns. Ideally, we can able to encode the n values in categorical variable into n-1 values. It means, each variable can be represented by the values of rest of the variables. Hence, we drop one of the variables, usual practice is to drop first variable created.

Hence, we give drop_first=True in argument of dummy variable creation, so that first variable can be ignored.

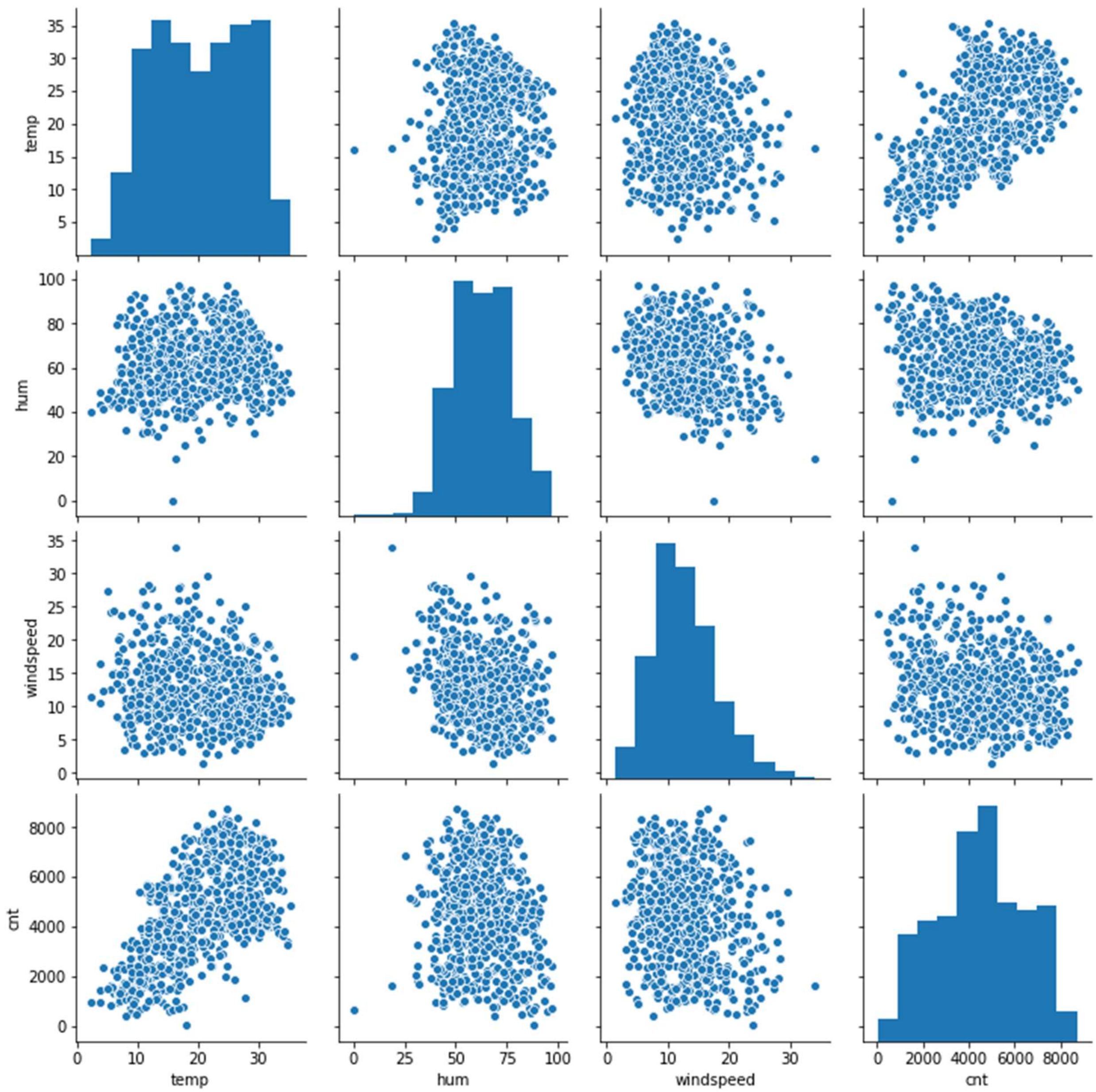
If you don't drop one of the column then your dummy variables will be correlated, this may adversely affect the models.

Ex: In Cities column we have values like ‘Delhi’, ‘Mumbai’ and ‘Bangalore’. When we convert these into dummies ‘Mumbai’ and ‘Bangalore’ will be populated with 1 and 2 respectively. If any rows have both ‘Mumbai’ and ‘Bangalore’ values as 0, then such row related to ‘Delhi’ city.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

By looking at below pair-plot we can deduce ‘temp’ variable has highest correlation with the ‘cnt’ target variable among the numerical variables.

	temp	atemp	hum	windspeed	cnt
temp	1.000000	0.991696	0.128565	-0.158186	0.627044
atemp	0.991696	1.000000	0.141512	-0.183876	0.630685
hum	0.128565	0.141512	1.000000	-0.248506	-0.098543
windspeed	-0.158186	-0.183876	-0.248506	1.000000	-0.235132
cnt	0.627044	0.630685	-0.098543	-0.235132	1.000000

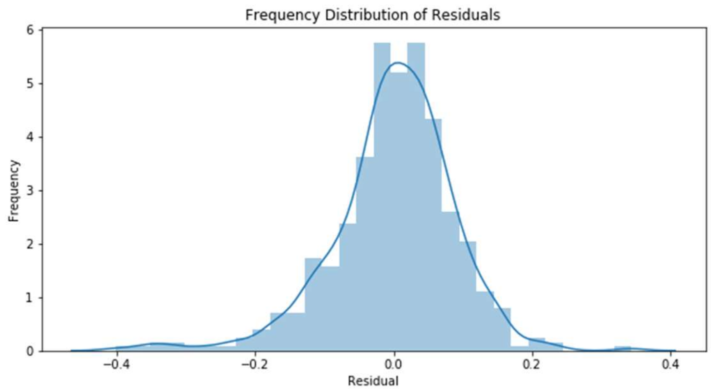


4. How did you validate the assumptions of Linear Regression after building the model on the training set?

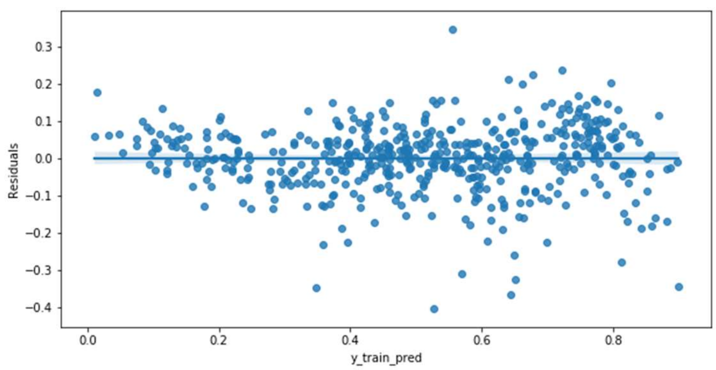
Assumptions of Linear Regression are as follows:

1. Error terms are normally distributed with mean zero.
2. Error terms are independent of each other.
3. Error terms have constant variance (homoscedasticity).

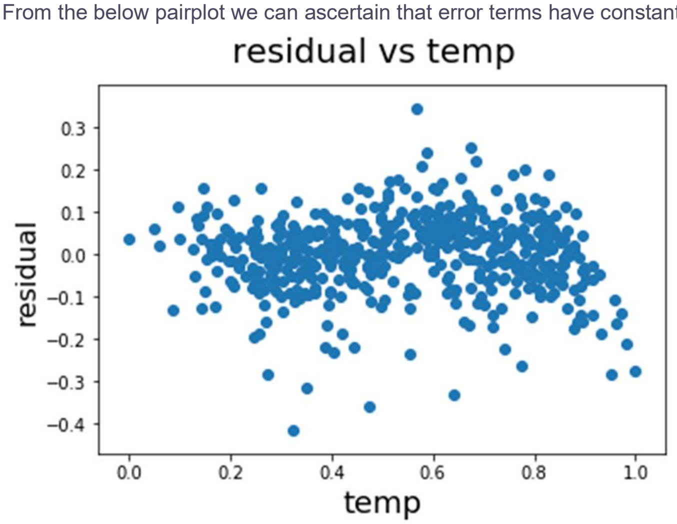
From the below residual distribution graph we can conclude that error terms are normally distributed and mean at zero.



From the below pairplot we can ascertain that error terms are independent of each other.



From the below pairplot we can ascertain that error terms have constant variance (homoscedasticity).



5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Below are the coefficients values after building the final model. Its clear that “temp”, “weathersit_Light Snow & Rain

” and “yr” variable are significantly explaining the demand for the shared bikes.

Coefficients of the variables as below:

- temp : 0.512675
- weathersit_Light Snow & Rain : -0.248516
- yr : 0.230700

index	Variables	Coefficient value
10	temp	0.512675
8	yr	0.230700
0	const	0.216679
3	season_Winter	0.111924
4	mnth_Sep	0.094023
2	season_Summer	0.068717
5	weekday_Monday	0.062701
9	workingday	0.053223
1	season_Spring	-0.044772
7	weathersit_Mist & Cloudy	-0.057310
11	hum	-0.150823
12	windspeed	-0.179875
6	weathersit_Light Snow & Rain	-0.248516

General Subjective Questions

1. Explain the linear regression algorithm in detail. (4 marks)

Linear Regression Algorithm is a machine learning algorithm based on supervised learning. Linear regression is a part of regression analysis. Regression analysis is a technique of predictive modelling that helps you to find out the relationship between Input and the target variable.

Regression analysis is used for three types of applications:

1. Finding out the effect of Input variables on Target variable.
2. Finding out the change in Target variable with respect to one or more input variable.
3. To find out upcoming trends.

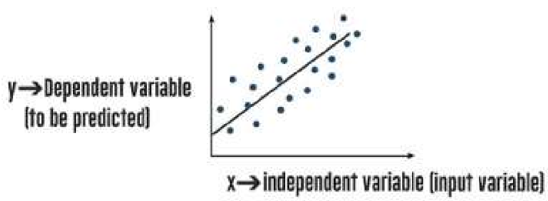
Here are the types of regressions:

1. Linear Regression
2. Multiple Linear Regression
3. Logistic Regression
4. Polynomial Regression

Linear regression is one of the very basic forms of machine learning where we train a model to predict the behavior of data based on some variables. As name suggests, two variables on x-axis and y-axis should be linearly correlated. Linear regression is based on the popular equation **"y = mx + c"**.

It assumes that there is a linear relationship between the dependent variable(y) and the predictor(s)/independent variable(x). In regression, we calculate the best fit line which describes the relationship between the independent and dependent variable. Regression is performed when the dependent variable is of continuous data type and Predictors or independent variables could be of any data type like continuous, nominal/categorical etc.

Regression method tries to find the best fit line which shows the relationship between the dependent variable and predictors with least error. In regression, the output/dependent variable is the function of an independent variable and the coefficient and the error term.



Ex: If we have dataset which contains information about the relationship between number of hours studies and marks obtained in some subject (ex: Machine Learning 1). Suppose this is our training data, and our goal here is to design a model which can predict the marks if number of hours student studies is provided. Using the training data, a regression line is created with minimum error. This linear equation is used to predict marks obtained for future data; our model should able to predict students marks with minimum error.

Regression is divided into two types as below:

1. Simple Linear Regression:
    a. It is used when the dependent variable is predicted using only one independent variable.
2. Multiple Linear Regression:
    a. It is used when the dependent variable is predicted using multiple independent variables.

Multiple Linear Regression Equation:

$$y_i=\beta_0+\beta_1 x_{i1}+\beta_2 x_{i2}+\ldots+\beta_p x_{ip}+\epsilon$$

where, for i=n observations:

yi = dependent variable

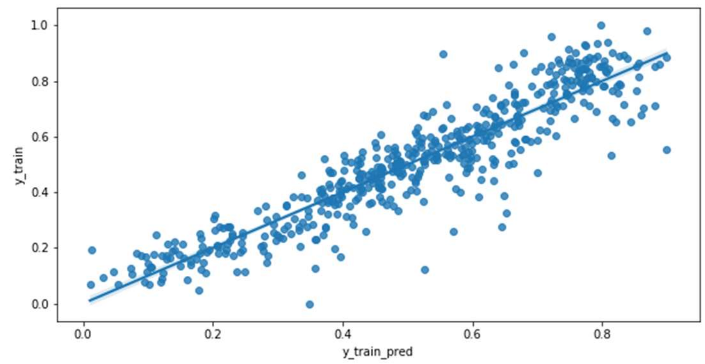
xi = explanatory variables

$\beta 0$ = y-intercept (constant term)

βp = slope coefficients for each explanatory variable

ϵ = the model's error term (also known as the residuals) .

The aim of regression analysis is to create a predicted line based on the training data we have acquired, which can then be used to confirm or deny the relationship between attributes. If the test is done over a long-time duration, extensive data can be collected, and the result can be evaluated more accurately. Below picture demonstrates the line which **best fits** the Bike Rental Analysis data.

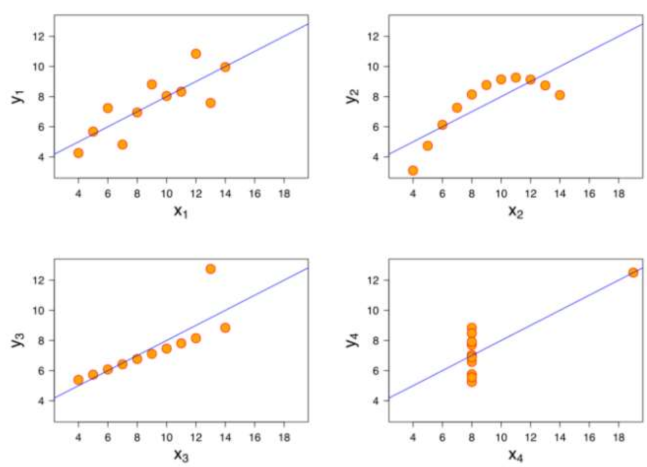


2. Explain the Anscombe's quartet in detail. (3 marks)

Anscombe's quartet explains that the different data sets can have similar statistical characteristics, however they have very different distribution and appear completely different when plotted on the graph.

Anscombe's quartet comprises four data sets as show below, it explains two important aspects.

1. Importance of graphing data before analyzing it.
2. The effect of outliers and other significant observations on statistical properties.



- Dataset I appear to have clean and well-fitting linear models.
- Dataset II is not distributed normally.
- In Dataset III the distribution is linear, but the calculated regression is thrown off by an outlier.
- Dataset IV shows that one outlier is enough to produce a high correlation coefficient.

Essentially, quartet explains the importance of visualization in Data Analysis, just by looking at the data reveals a lot of the structure and a clear picture of the dataset.

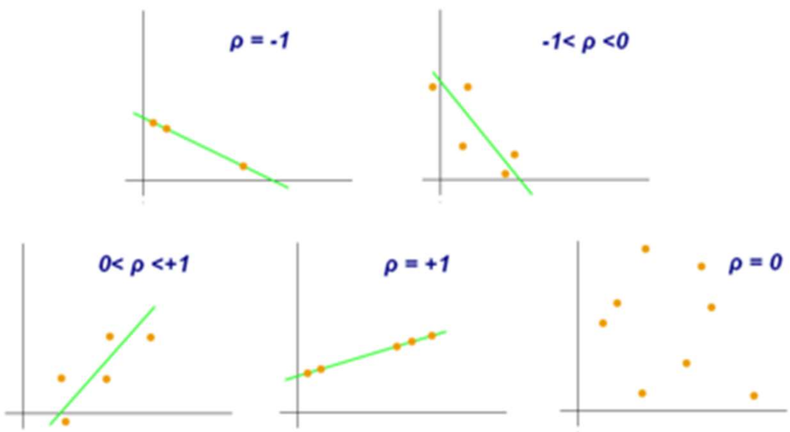
3. What is Pearson's R? (3 marks)

As per the definition, The Pearson's R is a statistic that measures linear correlation between two variables X and Y. In other words, Pearson's correlation coefficient is a **statistical measure how string relationship between two variables**.

The **is Pearson's R** formula return a value between -1 and 1, where:

- 1 indicates a strong positive relationship.
- -1 indicates a strong negative relationship.
- A result of zero indicates no relationship at all.

Graphs demonstrating a correlation of -1, 0 and +1.



4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling? (3 marks)

Feature Scaling is basically transformation of the data such that features are within a specific range ex: 0 to 1. Essentially, it is the process used to standardize the range of features of data. In practice its very common to see range of values for the data, it becomes very important stage in data preprocessing as part of machine learning algorithms building methods.

If we ignore the scaling, it has adverse effect on the ML models building process. Most of the times real world datasets contains features strongly varying in magnitudes and ranges. ML algorithms usually find distance between two data points, if we don't consider scaling this will create problem. Ex: datasets with 1 meter or and 1000 mm, high magnitude will weigh lot more in distance calculation than the low magnitude features. To mitigate these issues, we need to consider feature scaling to same level of magnitudes.

Normalization: This technique re-scales a feature or observation value with distribution value between 0 and 1. Generally it is used when you know that the distribution of your data does not follow a Gaussian distribution.

$$X_{new} = \frac{X_i - \min(X)}{\max(x) - \min(X)}$$

Standardization: It is a very effective technique which re-scales a feature value so that it has distribution with 0 mean value and variance equals to 1. Unlike normalization, standardization doesn't have a bounding range. Essentially if data has an outlier, they will not be affected by standardization.

$$X_{new} = \frac{X_i - X_{mean}}{\text{Standard Deviation}}$$

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen? (3 marks)

Variance Inflation Factor (VIF) – It's an index that provides the measure of how much variance of estimated regression coefficient increases due to collinearity. To calculate VIF value we fit regression model between the independent variables.

- VIF = $1/(1 – R^2)$
- Where R^2 is the R-square value of that independent variable.
- When R^2 reaches 1, VIF reaches to Infinity.

If an independent variable can be explained perfectly by other independent variables, then it will have perfect correlation and its R-squared value will be equal to 1.

So, VIF = 1/(1-1)

VIF = 1/0, which result in **VIF =** "**infinity**".

Higher the values of VIF signify that it is difficult to assess accurately the contribution of predictors variables to the model.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression. (3 marks)

A Q–Q plot is a plot of the quantiles of two distributions against each other, or a plot based on estimates of the quantiles. The pattern of points in the plot is used to compare the two distributions. It is also explained as a graphical technique for determining if two data sets come from populations with a common distribution.

A q-q plot is a plot of the quantiles of the first data set against the quantiles of the second data set. Ex: When we say 0.25 (25%) quantile, it means at this quantile point 25% of points fall below and 75% fall above that point.

We also draw a 45-degree reference line, If the two sets come from a population with the same distribution, then the points should fall approximately along this reference line. The greater the departure from this reference line, the higher the proof for the conclusion that the two datasets have come from populations with different distributions.

Advantages of Q-Q plot.

1. If two data sets have come from populations with a common distribution.
2. And they have common location and scale.
3. Whether they have similar distributional shapes.
4. If they have similar tail behavior.